

Playing Close to the Vest: Competitive Information Propagation in Partially Observed Dual-Population Mean-Field Games

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Abstract

Mean-field games (MFGs) have emerged as a foundational methodology for modeling information propagation within social networks, effectively bridging micro-level individual user decisions and macro-level emergent diffusion dynamics. However, a critical limitation of conventional MFG models is their reliance on the assumption of homogeneity, which fails to account for the inherent heterogeneity in user behaviors. We propose Cyclic Partially Observable Mean-Field Reinforcement Learning (CPO-MFRL), for Dual-Population Partially Observable Mean-Field Games (DPPO-MFG), a scalable learning framework that alternates policy evaluation and improvement across two interacting populations. Our method leverages a dual population policy-aware modeling layer (DP-PAM) and a lifted MFG with constrained policy space to decouple cross-population dependencies while preserving equilibrium structure. To address partial observability, we propose an encoder-decoder agent to reconstruct states and action during training. We further establish the existence of constrained Nash equilibria in the lifted setting and prove that CPO-MFRL converges to an equilibrium under partial observation by estimating mean actions via Dirichlet aggregation and derive high-probability bounds on action and Q-function discrepancies, which translate into uniform control of return gaps, with sample complexity scaling characterized by the number of agents, action dimension, and confidence level. Empirically, CPO-MFRL achieves rapid, stable convergence and superior performance on DPPO-MFG

benchmarks, i.e., the metric quantifying deviations from equilibrium strategies, by 61.8% on average compared to baselines, demonstrating robustness to observation noise and population heterogeneity. Together, these results show that cyclic population-wise updates, coupled with policy-aware modeling and statistically controlled mean-action estimation, provide a principled and effective route to learning equilibria in DPPO-MFGs.

CCS Concepts

• Computing methodologies → Multi-agent reinforcement learning.

Keywords

Information propagation, partial observation, mean-field game, reinforcement learning.

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1 Introduction

Social networks serve as the principal medium for multimedia delivery, facilitating global-scale engagement among billions of users through dynamic interactions, content dissemination, and information reprocessing. Within this ecosystem, information propagation strategies are essential for optimizing both the reach and impact of multimedia content. The effectiveness of propagation strategies is contingent upon the structural topology and user behavioral dynamics inherent to social networks. The central challenge in the design of efficient propagation strategies resides in the comprehensive understanding of collective user behavior, where individual decisions (e.g., re-posting, liking, or disregarding content) are interdependent and collectively shape macro-level information diffusion.

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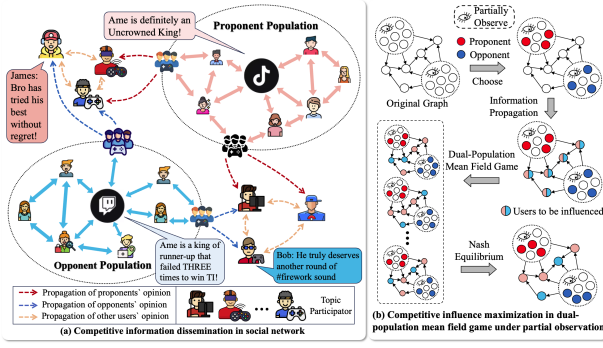


Figure 1: Competitive propagation in dual population mean field games.

Mean-Field Games (MFGs) model strategic interactions in large populations and extend to multi-population, partially-observed settings where distinct groups interact through aggregate effects, as in markets or security. In social networks, MFGs have been applied to opinion evolution [20], herding behaviors [7], and information propagation [28]. However, current MFG models assume user homogeneity, neglecting the heterogeneity of user preferences in content consumption and propagation. Social networks are inherently heterogeneous, with users clustering into groups based on shared interests, culture, or ideology, leading to diverse behaviors and inter-group interactions like competition, collaboration, or polarization that crucially influence information diffusion.

Competitive propagation, where adversarial groups spread conflicting information like political narratives or rival products, is a key form of inter-group interaction. We introduce Dual-Population Partially Observable Mean-Field Games (DPPO-MFG) to model competitive propagation between adversarial groups. This framework accounts for user heterogeneity and partial observability, addressing fundamental theoretical challenges regarding equilibrium existence and stability in complex information diffusion environments.

Conventional approaches [1, 2] require full environmental knowledge and solve complex differential equations, yielding closed-form solutions only in limited cases like linear-quadratic models with high computational costs that limit social network scalability.

Recent Deep Reinforcement Learning (DRL) approaches effectively solve complex MFGs by approximating high-dimensional distributions and adapting to dynamic environments [11, 26, 32, 34]. However, adapting DRL to competitive information propagation in DPPO-MFG remains an open challenge, despite DRL’s strong applicability to partially observable problems.

In light of the above challenges, we design a DPPO-MFG for competitive propagation and propose a cyclic deep reinforcement learning framework to achieve equilibrium. The **contributions** of this paper are summarized as follows:

Dual-Population Mean-Field Game for Competitive Propagation. We propose a DPPO-MFG to model competitive information propagation between two antagonistic populations. The framework rigorously proves the existence and uniqueness of a Nash equilibrium for this strategic interaction under partial observation.

Cyclic Partially Observable Mean-Field Reinforcement Learning (CPO-MFRL). We develop CPO-MFRL, a dual-agent architecture where sequentially interacting agents represent competing populations in DPPO-MFG. Using graph autoencoders to process network states, CPO-MFRL is proven to converge to a constrained Nash Equilibrium.

Practical Evaluation. We conduct experiments on 4 real-world social network datasets i.e. congress-Twitter [10], soc-sign-bitcoin-otc [14], soc-sign-bitcoin-alpha[14], ego-Facebook[18], and compared with 3 SOTA baselines: M3RL[25], EBPG[24], SMFU[16]. Our method achieves up to 61.8% improvement in terms of propagation coverage and 50.63% reduction in terms of exploitability (i.e., the metric quantifying deviations from equilibrium strategies) compared to SOTA baselines.

2 Related Work

Competitive Information Propagation: Competitive information propagation is a critical topic in social media analysis. The existing works in the domain can be categorized into two cases: known opponent strategies and unknown opponent strategies. Bharathi et al. [8] were the first to define the competitive information propagation problem within a game-theoretic framework. In recent years, various machine learning-based approaches have been employed to tackle this problem, as detailed below. Lin et al. [17] address both known and unknown opponent strategies in the competitive propagation problem, using Q-learning for known strategies and game-theoretic methods for Nash equilibrium. Ali et al. [3] extend this work to a Time-Constrained competitive propagation scenario, where groups design strategies within a limited time to maximize influence spread, using Q-learning with a time-varying linear threshold (LT) model. Yang et al. [31] introduce an RL-based framework for multi-round, multi-party influence maximization, while Ali et al. [5] use DQN to solve competitive propagation in unknown topology networks, outperforming heuristic methods. He et al. [12] investigate competitive opinion maximization with negatively weighted edges, using Q-learning for seed node selection without prior knowledge of opponent strategies. Ali et al. [6] proposed the DRL-EMB framework, combining node2vec and DQN to model dynamic social networks. Additionally, Ali et al. [4] improved competitive LT models with transfer learning, reducing computational time.

Multi-Population Mean-Field Game. Mean Field Games focus on modeling large-player games, examining how individual behaviors affect system dynamics. Recent advancements in MFG research, particularly with reinforcement learning, have addressed the limitations of traditional equilibrium computation methods. Sarah Perrin et al.[23] reformulated group aggregation as an MFG and introduced the Flock’n RL algorithm, combining deep RL and normalizing flows to solve multi-group aggregation problems and identify Nash equilibria. Mathieu Laurière et al.[15] proposed two D-AFP and D-MOMD to improve the efficiency of equilibrium computation and nonlinear approximation, achieving superior performance across MFG scenarios. Gökçe Dayankılı et al.[9] introduce a stylized model for carbon emissions with multiple mean field populations and one major player per population, considering network interactions, moreover, they provide analytical solutions for

minor and major players' controls at equilibrium based on Pontryagin's maximum principle. Wang et al. [30] formulate the cooperative attack-defense evolution of large-scale agents in high-dimensional environments as an MPHD-MFG, reducing communication frequency and computational complexity and they propose ECA-Net to solve the MPHD-MFG. Luigi De Pascale et al. [22] propose a multi-population mean field game model with quadratic Hamiltonian and non-local interactions, resulting in a system of coupled PDEs. The authors introduce Eulerian and Lagrangian variational formulations for the model. They prove the existence of weak solutions in different settings. A numerical method based on a Sinkhorn-like scheme is developed, and numerical results show different behaviors according to the non-local interaction nature, providing insights into multi-population MFGs.

Partial Observation. Social network analysis fundamentally relies on complete and accurate data on ties between actors to model structural properties. However, the empirical reality of sociological research often necessitates the use of partially observed networks. This approach, while introducing specific methodological challenges, remains a critical and pragmatic tool for investigating social structures, particularly in large-scale, hidden, or hard-to-reach populations. Georgios Papoudakis et al. [21] address agent modelling under partial observability by proposing an encoder-decoder method that learns representations of other agents' behaviours using only the controlled agent's local observations and actions. Andrew Wang et al. [29] introduce "Believer," a novel deep reinforcement learning method for partial observability that leverages full state access during training to learn compact belief representations for deployment, which outperforms standard recurrent policies in complex, partially observable domains.

3 Competitive Propagation as Dual-Population Partial Observation Mean-Field Games

The competitive propagation problem considers two adversarial populations: the proponent population and the opponent population, competing within a social network.

3.1 Graph Model for Competitive Propagation

The social network can be modeled as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the node set $\mathcal{V}$ represents users and the edge set $\mathcal{E}$ represents relationships between nodes. Each edge $e = u, v \in \mathcal{E}$ is associated with an activation weight $w_{u,v}$, indicating the probability that node v is influenced by the content propagated by node u . The proponent population (PP) and opponent population (OP) in competitive propagation can be written as sub-graphs of the social network, denoted by $\mathcal{G}^P = (\mathcal{V}^P, \mathcal{E}^P) \subset \mathcal{G}$ and $\mathcal{G}^O = (\mathcal{V}^O, \mathcal{E}^O) \subset \mathcal{G}$, respectively. Besides PP and OP, we also introduce the environmental population (EG), denoted by $\mathcal{G}^E = (\mathcal{V}^E, \mathcal{E}^E) \subset \mathcal{G}$, which contains target users of PP and OP to be influenced by information propagation. Collectively, we have $\mathcal{G} = \mathcal{G}^P \cup \mathcal{G}^O \cup \mathcal{G}^E$.

Let $\mathcal{W} = P, O$ index the proponent and opponent populations. Each designs a propagation strategy, strategically selecting a limited set of seed nodes from within its own members to maximize influence amidst competition. This selection occurs over discrete epochs. In epoch t , the seeds chosen for population w are $\mathcal{U}_t^w \subseteq \mathcal{V}^w$,

with $|\mathcal{U}_t^w| \leq K$ as a budget-driven limit. The information diffusion process then initiates from these seeds according to a defined model.

Next, we briefly introduce the diffusion model in the context of competitive propagation. This diffusion model for competitive propagation defines an "activated node" as one influenced by information. Seed nodes are initially activated and can influence neighbors. Influence attempts fail on nodes from rival populations, while environmental nodes are influenced with a probability equal to the edge weight. Let Λ_t^w be the nodes activated by population w at time t . Each environmental node v tracks the degree of influence received from Λ_t^w . The influence of population w received at environmental node v can be calculated by $F^w = \sum_{u \in \Lambda_t^w} w_{u,v}$. Let $a_{v,t} \in \{-1, 0, 1\}$ denote the activation state of environmental node v at time t , its value is determined by the following rule

$$a_{v,t} = \begin{cases} 1, & F_{v,t}^P > F_t^O \text{ and } F_t^P \geq \text{Thr}_v^P, \\ -1, & F_t^O > F_t^P \text{ and } F_t^O \geq \text{Thr}_v^O, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where $a_{v,t} = 1$ indicates that node v is activated by PP, and $a_{v,t} = -1$ indicates the environment node v is activated by OP, and $a_{v,t} = 0$ denote the node is not influenced. Thr_v^P and Thr_v^O denote the activation threshold for proponent and opponent opinion, respectively.

The reward of the seed selection reflects the number and quality of activated environmental nodes. Note the the current activated nodes Λ_t^w are resulted from a sequence of seed selection decision $\{\mathcal{U}_\tau^w\}_{\tau=0}^t$. To correctly reflect the reward achieved by the seed nodes in $\mathcal{U}_t^w$, we record activated nodes in the previous epoch $t-1$, Λ_{t-1}^w , which helps to isolate the marginal propagation impact by removing the effect of nodes activated in the preceding time step. Then, the reward of seed selection action in $\mathcal{U}_t^w$ is defined as

$$r_t^w = \alpha(|\Lambda_t^w| - |\Lambda_{t-1}^w|) + \beta \left(\sum_{v \in \Lambda_t^w \text{ \& } v \notin \Lambda_{t-1}^w} d_v - \sum_{v \in \Lambda_{t-1}^w \text{ \& } v \notin \Lambda_t^w} d_v \right) \quad (2)$$

where d_v is the degree of node v , α and β are importance weights. The first term measures the changes in the number of activated nodes, and the second term measures the changes in the impact of activated nodes.

3.2 Dual-Population Partially Observable Mean-Field Games

Given the vast scale of social networks, individual user behavior is negligible, whereas collective dynamics captured by the mean field are significant. This motivates formulating information propagation as a mean-field game. However, traditional mean-field games assume user homogeneity, whereas competitive propagation involves heterogeneous populations, thus motivating our Dual-Population Partially Observable Mean-Field Game (DPPO-MFG).

DPPO-MFG considers users in PP and OP collectively under the partial observation setup, wherein users within the same population are homogeneous and users from different populations are heterogeneous. The state, observation and action spaces for population $w \in \mathcal{W}$ are denoted as $\mathcal{S}^w$, $\mathcal{O}^w$ and $\mathcal{A}^w$, respectively, both of which are finite. DPPO-MFG evolves over discrete time steps, and we let $\mathcal{T} = \{0, 1, \dots, T\}$ denote the time index set. At time $t \in \mathcal{T}$, let $s_{i,t}^w \in \mathcal{S}^w$ denote the abstract representation of discrete

features (e.g., account information, number of fans, seed indicator, activation state, etc.) of a user $i \in \mathcal{V}^w$ and $o_{i,t}^w \in \mathcal{O}^w$ denote the observation of a user $i \in \mathcal{V}^w$ in population $w \in \mathcal{W}$. The overall state of a population is captured by the empirical state distribution, which is a statistical measure that captures the proportion of agents occupying each state within a population. The empirical state distribution for population w at time t can be obtained by $\mu_t^w(s) = \frac{1}{|\mathcal{V}^w|} \sum_{i=1}^{|\mathcal{V}^w|} \mathbf{1}(s_{i,t}^w = s)$, where $\mathbf{1}(\cdot)$ is an indicator function and s is an abstract representation of a feature vector composed of a finite set of specific discrete features. The empirical state distribution is a probability measure on $\mathcal{S}^w$, i.e., $\mu_t^w(s) \in \Delta(\mathcal{S}^w)$ where $\Delta(\mathcal{S}^w)$ is the space of probability measures. Based on the empirical state distribution, the population can be modeled as a representative and collapses all other members into a distribution, which is called the *mean field*. For population w , its mean-field is denoted by $m_t^w(s) \in \Delta(\mathcal{S}^w)$, which is defined as the limit of the empirical distributions as the number of users in population w become infinity, i.e., $m_t^w(s) = \lim_{|\mathcal{V}^w| \rightarrow \infty} \mu_t^w(s)$.

With the mean-field, we focus on the representative of populations, and omit the user index i henceforth. The policy of representative w is denoted by $\pi^w : \mathcal{O}^w \rightarrow \Delta(\mathcal{A}^w)$. Based on the observed states, representative takes action $a_t^w \sim \pi^w(o_t^w)$. The representative w receives a reward $r_t^w(s_t^w, a_t^w, m_t^w, m_t^{w^-})$ (e.g., in the form of Eqn.(2)) after implementing the action. The reward depends not only on the state and action, but also on the mean-field of both the proponent and the opponent population.

When the epoch t ends, the state of representative w transitions to a new state according to a transition function: $\mathcal{P}^w : \mathcal{S}^w \times \mathcal{A}^w \times \Delta(\mathcal{S}^w) \rightarrow \Delta(\mathcal{S}^w)$. The new state is denoted by $s_{t+1}^w \sim \mathcal{P}^w(s_t^w, a_t^w)$. The evolution of the mean field is governed by the Fokker-Planck equation, which can be written as $m_{t+1}^w(s_{t+1}^w) = \mathbb{E}[m_t^w(s_t^w)(\pi^w(a_t^w|s_t^w))\mathcal{P}^w(s_{t+1}^w|a_t^w, s_t^w)]$. Given policy π^w , let m_t^{w,π^w} be the mean-field of population w reached by implementing policy π^w . The value function of representative w is defined by:

$$V_t^w(\pi^w, m_t^{w,\pi^w}, m_t^{w^-, \pi^{w^-}}) = \mathbb{E}_{a_t^w \sim \pi^w, s_{t+1}^w \sim \mathcal{P}^w} \left[\sum_{\tau=t}^T r_\tau^w(s_\tau^w, a_\tau^w, m_\tau^{w,\pi^w}, m_\tau^{w^-, \pi^{w^-}}) \right].$$

The value function represents the expected cumulative rewards over the time horizon. In DPPO-MFG, each population designs a propagation strategy to maximize its value function with the competitive behaviors exhibited by its rival population.

Definition 1. DPPO-MFG for Competitive Propagation: The PP and OP in competitive propagation form a DPPO-MFG, where each population $w \in \mathcal{W}$ aims to find the optimal propagation policy π^{w*} that maximizes its value function while playing against the optimal propagation policy π^{w^*} of its rival population w^- , i.e., $\pi^{w*} = \arg \max_{\pi^w} V_t^w(\pi^w, m_t^{w,\pi^w}, m_t^{w^-, \pi^{w^*}})$.

Let us define the exploitability, which measures the distance of the current policy to an equilibrium:

Definition 2. Exploitability:

$$\Phi(\{\pi^w\}_{w \in \mathcal{W}}) = \sum_{w \in \mathcal{W}} \Phi^w(\pi^w) \quad (3)$$

where $\Phi^w(\pi^w) = \max_{\pi^{w^*}} J^w(\pi^{w^*}, \mu^\pi) - J^w(\pi^w, \mu^\pi)$

In the above equation, $\mu^\pi = \{\mu^{w,\pi^w}\}_{w \in \mathcal{W}}$ and $J^w(\pi^w, \mu^\pi) = V_t^w(\pi^w, \mu_t^{w,\pi^w}, \mu_t^{w^-, \pi^{w^-}})|_{t=0}$.

4 Cyclic Partially Observable Mean-Field Reinforcement Learning for DPPO-MFG

To address the lack of prespecified dynamics in real-world settings, we propose Cyclic Partially Observable Mean-Field Reinforcement Learning (CPO-MFRL), a model-free DRL framework that learns DPPO-MFG equilibria without prior environmental knowledge. Figure 2 provides an overall framework for CPO-MFRL. Our code and detailed proof process is available at <https://anonymous.4open.science/r/WWW2026-152F>.

4.1 Architecture of CPO-MFRL Agents

4.1.1 Dual-agent for dual-population. The DPPO-MFG framework is compatible with Markov Decision Processes (MDP), enabling natural integration with DRL. For handling two competing populations, our CPO-MFRL employs a dual-agent architecture where each agent represents one population. The agent for population w can be written as a parameterized policy $\pi^w(o^w; \Phi^w)$ where Φ^w denotes trainable parameters in the form of deep neural networks. For each decision epoch $t \in \mathcal{T}$, the agent for population w observes s_t^w and obtains an action $a_t^w = \pi^w(o_t^w; \Phi^w)$ (i.e., the seed selection decision $\mathcal{U}_t^w$) from the agent. μ^w shows the empirical state distribution of the population w . At the end of the decision epoch t , the agent receives a r_t^w (Eqn. (2)), and the state transitions to $s_{t+1}^w \sim \mathcal{P}^w(s_t^w, a_t^w)$. The four-tuple $(s_t^w, o_t^w, a_t^w, r_t^w, s_{t+1}^w, o_{t+1}^w)$ is the experience of decision epoch t and is stored in the replay memory $\mathcal{D}_t^w$. The goal of agent w in each decision epoch t is to maximize the total reward starting from t : $R_t^w = \sum_{\tau=t}^T \gamma^{\tau-t} r_\tau^w$, where $0 \leq \gamma \leq 1$ is the time discount factor. Agents do not directly observe the mean fields m_t^w but instead learn them as parameters within the transition and reward functions during DRL. The two agents operate independently, with their interconnection emerging solely through the competitive propagation dynamics.

4.1.2 Graph autoencoder for graph-structured social network data. We employ a graph autoencoder with Graph Attention Network (GAT) encoders to model both the structural topology of graphs and node-level attributes in social networks. Each agent w uses a GAT encoder that inputs the feature matrix $\mathbf{X}^w = \{\mathbf{x}_v\}_{v \in \{\mathcal{V}^w \cup \mathcal{V}^E\}}$, where $\mathbf{x}_v \in \mathbb{R}^d$ represents node features for population w and environmental nodes.

GAT consists of multiple layers. The k -th layer computes normalized attention coefficients $\alpha_{ij}^{(k)} = \exp(e_{ij}^{(k)}) / \sum_{l \in \mathcal{N}(i)} \exp(e_{il}^{(k)})$, with $e_{ij}^{(k)} = \text{Sigmoid}(\mathbf{v}_s^{(k)T} \sigma(\mathbf{W}^{(k)} \mathbf{t}_i^{(k-1)}) + \mathbf{v}_r^{(k)T} \sigma(\mathbf{W}^{(k)} \mathbf{t}_j^{(k-1)}))$, where σ is the activation function, and $\mathbf{v}_s^{(k)}, \mathbf{v}_r^{(k)} \in \mathbb{R}^{d^{(k)}}$, $\mathbf{W}^{(k)} \in \mathbb{R}^{d^{(k)} \times d^{(k-1)}}$ are trainable parameters. Node embeddings are updated as $\mathbf{t}_i^{(k)} = \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(k)} \sigma(\mathbf{W}^{(k)} \mathbf{t}_j^{(k-1)})$.

The decoder also utilized GAT architecture. Each graph attention layer attempts to reverse the process of its corresponding encoder layer. It computes coefficients $\hat{\alpha}_{ij}^{(k)} = \exp(\hat{e}_{ij}^{(k)}) / \sum_{l \in \mathcal{N}(i)} \exp(\hat{e}_{il}^{(k)})$, where $\hat{e}_{ij}^{(k)} = \text{Sigmoid}(\hat{\mathbf{v}}_s^{(k)T} \sigma(\hat{\mathbf{W}}^{(k)} \hat{\mathbf{t}}_i^{(k)}) + \hat{\mathbf{v}}_r^{(k)T} \sigma(\hat{\mathbf{W}}^{(k)} \hat{\mathbf{t}}_j^{(k)}))$, and reconstructs embeddings as $\hat{\mathbf{t}}_i^{(k-1)} = \sum_{j \in \mathcal{N}(i)} \hat{\alpha}_{ij}^{(k)} \sigma(\hat{\mathbf{W}}^{(k)} \hat{\mathbf{t}}_j^{(k)})$.

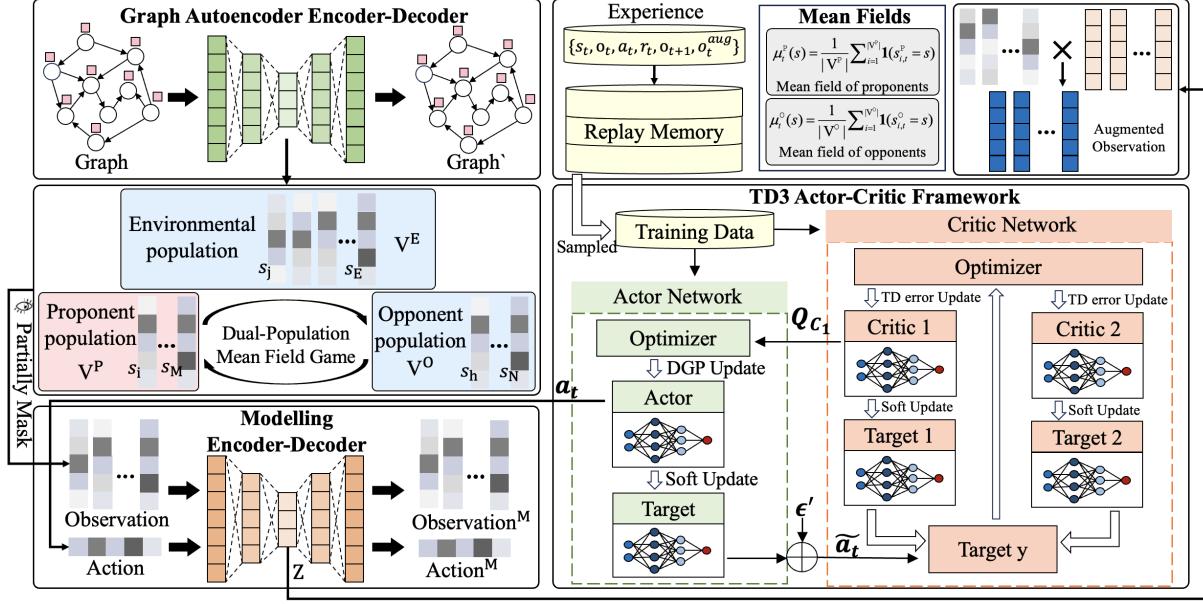


Figure 2: Framework of Cyclic Partially Observable Mean-Field Reinforcement Learning for DPPO-MFG.

Training minimizes the total loss $\mathcal{L}^T = \mathcal{L}^F + \lambda \cdot \mathcal{L}^A$, where $\mathcal{L}^F = \frac{1}{|\mathcal{V}^w \cup \mathcal{V}^E|} \sum_{i \in \{\mathcal{V}^w \cup \mathcal{V}^E\}} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$ is MSE for node feature reconstruction, $\mathcal{L}^A = -\frac{1}{|\mathcal{V}^w \cup \mathcal{V}^E|} \sum_{i \in \{\mathcal{V}^w \cup \mathcal{V}^E\}} \sum_{j \in \mathcal{N}(i)} \log \left(\frac{1}{1 + \exp(-t_i^T t_j)} \right)$ is binary cross-entropy for adjacency reconstruction, with λ as a weight parameter.

4.1.3 Agent Modelling under Partial Observability. This method employs an encoder-decoder architecture to learn compact opponent representations from the controlled agent's local observations and action history. After training via reconstruction, the decoder is discarded. The retained encoder conditions the agent's policy on these representations, with both components optimized concurrently during reinforcement learning.

The modelled agent's trajectories is to use a recurrent encoding model, that we denote as $f_w : \tau_1 \rightarrow Z$, with parameters w , to learn the relationship between the local trajectory of the controlled agent and the latent variables. Specifically, we learn the function that relates the modelled agent's trajectory to the latent variables with an encoder that only depends on local information of the controlled agent. Since during execution only the encoder is required to generate the latent variables of the modelled agent, this approach removes the assumption that access to the modelled agent's observations and actions is available during execution. At each time step t , the recurrent encoder network generates an embedding z_t , which is conditioned on the information of the agent under control (o_t^1, a_{t-1}^1) , until time step t . At each time step t the parametric decoder learns to reconstruct the modelled agent's observation and action $(o_t^1$ and $a_t^1)$ conditioned on the embedding z_t , which helps our strategy to constantly track the latest state change of the environment. $\mathcal{L}_{PO} = \frac{1}{H} \sum_{t=1}^H \left[(f_u^o(z_t) - o_t^1)^2 - \log f_u^\pi(a_t^1 | z_t) \right]$, where $z_t = f_w(o_{1,t}^1, a_{1,t-1}^1)$.

4.1.4 Actor-critic framework for handling action space explosion. The seed node selection is, in essence, a combinatorial optimization problem which has a high-dimensional discrete action space that expands exponentially as the number of candidate nodes increases. This renders traditional RL approaches, e.g., deep Q-learning [19], limited in implementation. To improve the learning efficiency, we transform the form of the action from a combinatorial set to a vector of real numbers $\mathbf{p}^w = \{p_v^w\}_{v \in \mathcal{V}^w} \in \mathbb{R}^{|\mathcal{V}^w|}$, where p_v^w denotes the probability of node that are selected as seeds. In such a case, we have a continuous action space that is linearly scalable. To learning with continuous action space, we employ actor-critic as our RL framework.

Each agent w maintains a parameterized policy $\pi^w(o^w, z^w | \phi^w)$, where ϕ^w is the parameters in form of deep neural networks.

Under the background of the partial observation setup, latent variables z augmented with the controlled agent's observation can be used in the RL optimised policy. Consider the augmented space $O_{aug}^1 = O^1 \times Z$, where O^1 is the original observation space of the controlled agent, and Z is the representation space about the agent's models. The advantage of learning the policy on O_{aug}^1 compared to O^1 is that the policy can specialise for different $z \in Z$. In our experiments we optimised the policy of the controlled agent using actor-critic framework, however, other RL algorithms could be used in its place. The input to the actor and critic are the local observation and the generated representation. We do not back-propagate the gradient from the actor-critic loss to the parameters of the encoder. We use different learning rates for optimising the parameters of the networks of RL and the encoder-decoder.

In actor-critic methods, the objective is to find the optimal policy $\pi^w(o^w | \phi^w)$, with parameters ϕ^w , which maximizes the expected return $J^w(\phi^w) = \mathbb{E}_{o^w \sim \mu^w, a^w \sim \pi^w} [R_t^w | t=0]$, where $R_t^w = \sum_{\tau=t}^T \gamma^{\tau-t} r_t^w$ is the long-term reward. The policy parameter of the policy is

updated by taking the gradient of the expected return $\nabla_{\phi^w} J^w(\phi^w)$, which can be obtained by the deep deterministic policy gradient algorithm:

$\nabla_{\phi^w} J^w(\phi^w) = \mathbb{E}_{o^w \sim \mu^w} [\nabla_{a^w} Q_1^w(o^w, z^w, a^w; \pi^w)]|_{a^w = \pi^w(o^w | \phi^w)} \cdot \nabla_{\phi^w} \pi^w(o^w | \phi^w)$ where $Q_1^w(o^w, z^w, a^w; \pi^w) = \mathbb{E}_{o^w \sim \mu^w, a_i^w \sim \pi^w} [R_{1,t}^w]$ is the Q-value of the 1st critic network when using the minimum value. The parameter of critic is updated by minimizing the Mean Square Bellman Error using the gradient descent strategy $\nabla_{\theta_i^w} \mathcal{L}(\theta_i^w)$: $\mathcal{L}(\theta_i^w) = \mathbb{E}_{(o^w, a^w, r^w, o^{w'}, d) \sim \mathcal{D}} \left[(Q_i^w(o^w, z^w, a^w; \pi^w) - y^w)^2 \right]$ where $y^w = r^w + \gamma(1-d) \cdot \min_{j=1,2} Q_j^w(o^{w'}, z^{w'}, \pi^{w'}(o^{w'}, z^{w'}) + \epsilon)$.

4.1.5 Cyclic sequential decision-making. Previous multi-agent systems research primarily assumed synchronous updates. While some studies (e.g. Chao Yu et al. [33]) explore asynchronous settings, they typically neglect game-theoretic interactions. In practical contexts, the decision-making of information propagation in the proponent and opponent population operates asynchronously rather than synchronously. For instance, propagation can be driven by an event-triggered mechanism, e.g., when certain users shift their opinion in response to the propagation of its rival. To model this inherent asynchrony, our method formalizes interactions within a dual-agent RL framework as a sequential process. Specifically, during each decision epoch t , the following sequence of operations occurs: (1) Agent $w \in \mathcal{W}$ observes the current state of the social network (e.g., user opinions and influence spread) and selects its seed nodes; (2) The rival agent w^- then observes the updated state of the social network, and subsequently determines its own seed selection. This process iterates cyclically, reflecting the dynamic interplay between competing agents.

Sequential decision-making in our framework reduces optimization complexity in high-dimensional spaces by decoupling strategy updates and anchoring them to observed opponent behavior, thereby mitigating non-stationarity caused by simultaneous updates. We formalize an event-triggered dual-population game that mirrors real-world social networks, where one population reacts only after observing new seed activations by its opponent—capturing the reactive, event-driven nature of competitive diffusion. This structure embeds adversarial learning within the sequential process: each agent observes the environment shaped by its rival's actions before responding. Such sequentiality reflects real-world dynamics, improves model tractability, and facilitates stable convergence to equilibrium policies.

4.2 Analysis of NE Convergence for CPO-MFRL under Partial Observation

We next analyze whether CPO-MFRL can converge to the NE of DPPO-MFG. Because any non-equilibrium strategy is exploitable, triggering adversarial responses and persistent instability. Firstly, we introduce the average situation without partial observation, DP-MFG. DP-MFG can be described by a collection of tuples $\mathbf{M} := \{(m_0^w, \mathcal{S}^w, \mathcal{A}^w, T, \mathcal{P}^w, r^w)_{w \in [W]}\} = \{M^w_{w \in [W]}\}$. Our key observation is that, the DP-MFG $\mathbf{M}$ can be lifted to a Mean-Field Markov Decision Process (MF-MDP) $\bar{M} := \{\bar{\mathcal{S}}, \bar{\mathcal{A}}, \bar{m}_0, T, \bar{\mathcal{P}}, \bar{r}\}$, by augmenting the original states and actions with the type index. The new state and action spaces are given by $\bar{\mathcal{S}} := \bigcup_{w \in [W]} \{\mathcal{S}^w \times \{w\}\}$, and

$\bar{\mathcal{A}} := \bigcup_{w \in [W]} \{\mathcal{A}^w \times \{w\}\}$. For policies in $\bar{M}$, we only consider $\Pi^\Psi := \{\pi \mid \forall w \in [W], \pi^w(a^w \mid s^w), \pi^w \in \Pi^w\}$, including all policies which only take actions with the same type as the states. For distribution in $\bar{M}$, we consider concatenating all mean field with normalization: $\bar{m} := [\frac{m^P}{2}, \frac{m^O}{2}]$.

For the function approximation setting, we assume dual transition function classes $\mathcal{M}^w$ are available for all population $w \in \mathcal{W}$, $\mathcal{M}^w \subset \{\{\mathcal{P}_t^w\}_{t \in [T]}, \mathcal{P}_t^w : \mathcal{S}_t^w \times \mathcal{A}_t^w \times \Delta(\mathcal{S}_t^w) \times \Delta(\mathcal{S}_t^w) \rightarrow \Delta(\mathcal{S}_{t+1}^w)\}$ is used to approximate the transition function for the population w . The DP-MFG transition function class $\mathcal{M}$ is defined by $\mathcal{M} \leftarrow \{\mathbf{M} := M^P \times M^O\}$, which is used to approximate the true model $\mathbf{M}^*$. In the following, we use $\bar{M}$ to denote the model class including MFG models converted from models in $\mathcal{M}$ using the similar method in Sec.4.2, and denote $\bar{M}^*$ to be the one converted from $\mathbf{M}^*$. Then we have the following assumptions:

Assumption A (Realizability). The true model $\mathbf{M}^* \in \mathcal{M}$.

Assumption B (Lipschitz Continuity for DP-MFG). For any $\mathbf{M} \in \mathcal{M}$, and two arbitrary policies $\pi, \tilde{\pi}$, the following stands $\forall w, t, s^w, a^w$: $\|\mathcal{P}_t^w(\cdot | s_t^w, a_t^w, m_t^\pi) - \mathcal{P}_t^w(\cdot | s_t^w, a_t^w, m_t^{\tilde{\pi}})\|_1 \leq \frac{L_T}{W} \|m_t^\pi - m_t^{\tilde{\pi}}\|_1$, $|r_t^w(s_t^w, a_t^w, m_t^\pi) - r_t^w(s_t^w, a_t^w, m_t^{\tilde{\pi}})| \leq \frac{L_T}{W} \|m_t^\pi - m_t^{\tilde{\pi}}\|_1$. Note that our Assump.B only requires Lipschitz continuity on feasible densities. Here, we introduce a normalization factor W given that $\|m_t^\pi - m_t^{\tilde{\pi}}\|_1 = \sum_{w \in [W]} \|m_t^{w,\pi} - m_t^{w,\tilde{\pi}}\|_1$.

To facilitate the analysis, we introduce the definition of NE in the DP-MFG under the constrained situation.

Definition 3 (Constrained NE in the Lifted Mean-Field MDP). When agents are constrained to only taking policies in the subset Π^Ψ , we call $\hat{\pi}_{\text{Cstr}}^{\text{NE}} \in \Pi^\Psi$ the ε -approximate Constrained Nash Equilibrium if $J_{\bar{M}}(\pi, \mu^{\pi_{\text{Cstr}}^{\text{NE}}}) \leq J_{\bar{M}}(\hat{\pi}_{\text{Cstr}}^{\text{NE}}, \mu^{\pi_{\text{Cstr}}^{\text{NE}}}) + \varepsilon, \forall \pi \in \Pi^\Psi$.

The following property reveals the connection between constrained NE in $\bar{M}$ and the NE in the original DP-MFG $\mathbf{M}$.

Proposition 1. Given a DP-MFG $\mathbf{M}$ and its lifted MF-MDP $\bar{M}$, we have: (1) an ε -constrained NE $\hat{\pi}_{\text{Cstr}}^{\text{NE}} \in \Pi^\Psi$ is a $(W\varepsilon)$ -NE in $\mathbf{M}$; (2) an ε -NE $\hat{\pi}^{\text{NE}}$ in $\mathbf{M}$ is an ε -constrained NE for $\bar{M}$.

The proof of Proposition 1 is given in the Appendix A.4. This proposition indicates that if an NE exists in the constrained policy, then an NE exists in the MF-MDP, which is lifted from the new constrained DP-MFG. Next, we establish the existence under the constrained condition.

Proposition 2. Given $\mathbf{M}$ satisfying Lipschitz continuity conditions in Assumption B, the MF-MDP $\bar{M}$ converted from $\mathbf{M}$ has at least one constrained NE satisfying $\forall \pi \in \Pi^\Psi, J_{\bar{M}}(\pi, \mu^{\pi_{\text{Cstr}}^{\text{NE}}}) \leq J_{\bar{M}}(\hat{\pi}_{\text{Cstr}}^{\text{NE}}, \mu^{\pi_{\text{Cstr}}^{\text{NE}}}) + \varepsilon$ with $\varepsilon = 0$.

The proof of Proposition 2 is given in Appendix A.5. According to Proposition 1 and Proposition 2, it can be concluded that the existence of constrained NE in $\bar{M}$ given the existence of NE in $\mathbf{M}$ by letting $\varepsilon \rightarrow 0$. This also suggests one can solve NE of DP-MFG by solving the constrained NE in its lifted MF-MDP.

Secondly, we introduce the particular situation under the partial observation which are the same as literature[27], and are followed by all the propositions and analyses below. Step by step, we state some assumptions:

Assumption C. For any i and j , $\lim_{t \rightarrow \infty} \tau_j^i(t) = \infty$ w.p.1.

Assumption D. $\mathcal{F}(t)$ is meant to represent the history of the algorithm up to, where t , and the measure-theoretic terminology that "a random variable Z is $\mathcal{F}(t)$ -measurable" means that Z is completely determined by the history represented by $\mathcal{F}(t)$:

- I) $x(0)$ is $\mathcal{F}(0)$ -measurable.
- II) For every i, j , and t , $w_i(t)$ is $\mathcal{F}(t+1)$ -measurable.
- III) For every i, j , and t , $\alpha_i(t)$ and $\tau_j^i(t)$ are $\mathcal{F}(t)$ -measurable.
- IV) For every i and t , we have $\mathbb{E}[w_i(t)|\mathcal{F}(t)] = 0$.
- V) For deterministic constants A and B , $\mathbb{E}[w_i^2(t)|\mathcal{F}(t)] \leq A + B \max_j \max_{\tau \leq t} |x_j(\tau)|^2$

Assumption E. The learning rates satisfy $0 \leq \alpha_i(t) < 1$.

Assumption F. F is assumed to be a mapping from $\mathbb{R}^n$ to itself which can be detailed in Tsitsiklis[27]:

- I) The mapping F is monotone, i.e. if $x \leq y$, then $F(x) \leq F(y)$.
- II) The mapping F is continuous.
- III) In the limit ($t \rightarrow \infty$), the mapping F is bounded in an interval $[x^* - D, x^* + D]$, where x^* is some arbitrary point.
- IV) If $e \in \mathbb{R}^n$ is the vector with all components equal to 1, and p is a positive scalar then, $F(x) - pe \leq F(x - pe) \leq F(x + pe) \leq F(x) + pe$.

Assumption G. Each action-value pair is visited infinitely often and the reward stays bounded.

Assumption H. The agents' policy is Greedy in the Limit with Infinite Exploration (GLIE).

Assumption I. The Nash equilibrium can be considered a global optimum or a saddle point in every stage of the stochastic game.

The assumptions of DPPO-MFG above are the validity of Mean-Field Theory, which dictates that, in a sufficiently large population, individual actions can be replaced by the mean-field. These assumptions are standard and widely accepted for applying the mean-field paradigm to real-world problems. In large-scale social networks, Assumptions C and G are naturally fulfilled due to the massive user volume and bounded graph structure of social networks, inherently providing sufficient data and finite returns. Assumptions A, B, H, and I are enforced via the integration of specialized components within the DPPO-MFG framework. Assumptions D, E, F are necessary conditions for the convergence of the TD3 algorithm's Bellman error minimization to a fixed point, which are widely used in many RL. Despite these assumptions not being perfectly met in practice, our method maintains strong performance on real-world benchmarks.

Proposition 3. *The DPMF mean action and the PODPMF mean action both satisfy $|\tilde{a}_{h,t}^w - \bar{a}_{h,t}^w| \leq \sqrt{\frac{1}{2n} \log \frac{2}{\delta}}$, as time $t \rightarrow \infty$, with probability $\geq \delta$, where n is the number of samples observed. $\bar{a}$ is the mean action obtained from the Eq.4 and $\tilde{a}$ is the mean action.*

Assuming that each agent is fully observable, the mean of important agents selected by our method is close to the true underlying global observable $\bar{a}_{h,t}^w$. The proof will be found in Appendix B.1.

Proposition 4. *When the Q-function is Lipschitz continuous (with constant M) with respect to mean actions, then the DPPO-MF Q-function will satisfy the following relationship:*

$$|\ddot{Q}_t^{w,\tilde{\pi}}(s_t^w, a_t^w, \tilde{a}_t^w; \pi^{NE}) - \ddot{Q}_t^{w,\tilde{\pi}}(s_t^w, a_t^w, \bar{a}_t^w; \pi^{NE})| \leq ML \log \frac{2}{\delta} \frac{1}{2n}$$

as $t \rightarrow \infty$ with probability $\geq (\delta)^{L-1}$, where $L = |A|$ and n is the number of samples.

When the Q-function is Lipschitz continuous with respect to the mean action, then the DPPO-MF Q-function $\ddot{Q}_t^{w,\tilde{\pi}}(s_t^w, a_t^w, \tilde{a}_t^w; \pi^{NE})$ is close to the MF Q-function $\ddot{Q}_t^{w,\tilde{\pi}}(s_t^w, a_t^w, \bar{a}_t^w; \pi^{NE})$. The proof will be found in Appendix B.2. According to Proposition 1~Proposition 4, we prove how the constrained NE we solve in Appendix B.3.

5 Experiments

5.1 Experimental Setups

Datasets and Baselines. We evaluate the proposed method on four real-world social network datasets: Twitter-Congress[10], Bitcoin-Network[14][13], Facebook[18]. Three SOTA baselines: M3RL [25], EBPG [24], SMFU [16], are employed for performance comparison to demonstrate the efficacy of our method. Detailed information about the dataset and baselines is provided in Appendix C.

Hyper-parameter Configurations. In the graph model for competitive propagation, the edge activation weight $w_{u,v}$ of graph $\mathcal{G}$ is defined as the inverse of the in-degree of node v . We detail the hyper-parameter in the Appendix D.

Evaluation Metric. We use two metrics: coverage and exploitability, to evaluate the performance of the proposed method and baseline. The coverage is used to evaluate the spread of the information. Mathematically, the coverage metric for the proponent and opponent populations is defined as $C_t^p = \frac{\sum_{i \in \mathcal{V}^E} 1(a_{i,t}=1)}{|\mathcal{V}^E|}$ and $C_t^o = \frac{\sum_{i \in \mathcal{V}^E} 1(a_{i,t}=-1)}{|\mathcal{V}^E|}$, respectively. The total coverage rate $C_t = C_t^p + C_t^o$ is used as the final metric. The exploitability reflects the closeness of the learned policy to the NE policy, which is defined in Eqn. (3).

5.2 Results and Discussions

Influence Coverage: Table 1 presents the performance of our proposed CPO-MFRL method against three baseline algorithms (M3RL, EBPG, and SMFU) across four distinct network datasets. Reflecting the differing scales of these networks, the seed budget was set to 2.00%, 4.00%, and 6.00% of the total nodes for the Congress-Twitter dataset, and to 0.50%, 1.00%, and 1.50% for the Bitcoin-alpha, Bitcoin-otc, and Facebook datasets. All results are reported as the average value from experiments run with three different random seeds, with standard deviations provided in parentheses. On the Congress-Twitter dataset, our method is not highly competitive, showing a little improvement at the 4.00% budget and matching the top performance at the 6.00% budget. This dataset's small scale, dense structure promoting rapid diffusion and its polarized, uniform clusters limit DPPO-MFG's advantage over heuristics. However, its superior capability becomes markedly evident on the larger and more complex networks. At the 0.50% seed budget, our method demonstrates substantial improvements in coverage compared to the second-best method: a 77.80% improvement on Bitcoin-alpha, a 107.44% improvement on Bitcoin-otc, and a 53.98% improvement on the Facebook network. The average relative improvement across these three datasets is 79.74%. These results underscore the scalability and efficiency of our approach, particularly in large-graph settings. This significant performance enhancement is attributable to the integration of graph neural networks within our DPPO-MFG framework, which enables CPO-MFRL to effectively capture the

	Congress-Twitter			Bitcoin-alpha			Bitcoin-otc			Facebook		
Budgets	2.00%	4.00%	6.00%	0.50%	1.00%	1.50%	0.50%	1.00%	1.50%	0.50%	1.00%	1.50%
M3RL	41.94(5.31)	59.85(5.93)	65.03(6.25)	27.55(3.31)	38.23(4.53)	46.69(4.85)	22.37(3.21)	32.33(4.32)	39.12(4.17)	21.34(7.91)	21.61(7.75)	36.23(8.97)
EBPG	47.38(5.37)	63.26(5.52)	71.29(5.41)	32.20(3.47)	49.39(3.92)	48.93(3.42)	23.84(4.14)	36.05(4.09)	45.32(5.07)	21.18(18.47)	33.62(20.57)	46.04(20.48)
SMFU	59.98(4.78)	76.90(5.31)	78.82(6.39)	29.38(3.25)	41.82(3.63)	56.56(5.03)	26.47(2.37)	39.65(2.54)	48.23(3.28)	26.23(9.02)	38.58(10.54)	54.18(10.32)
CPO-MFRL	62.14(6.39)	77.84(6.53)	79.01(6.83)	57.33(2.87)	72.82(2.91)	76.51(3.39)	54.91(5.29)	69.64(5.96)	75.32(6.91)	40.39(10.18)	52.92(10.08)	59.95(11.49)

Table 1: Coverage achieved by CPO-MFRL and baselines on evaluated datasets.

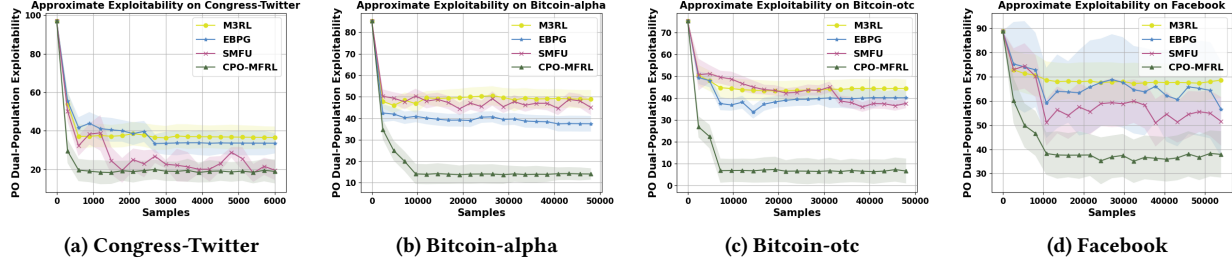


Figure 3: Evolution of exploitability metric during learning.

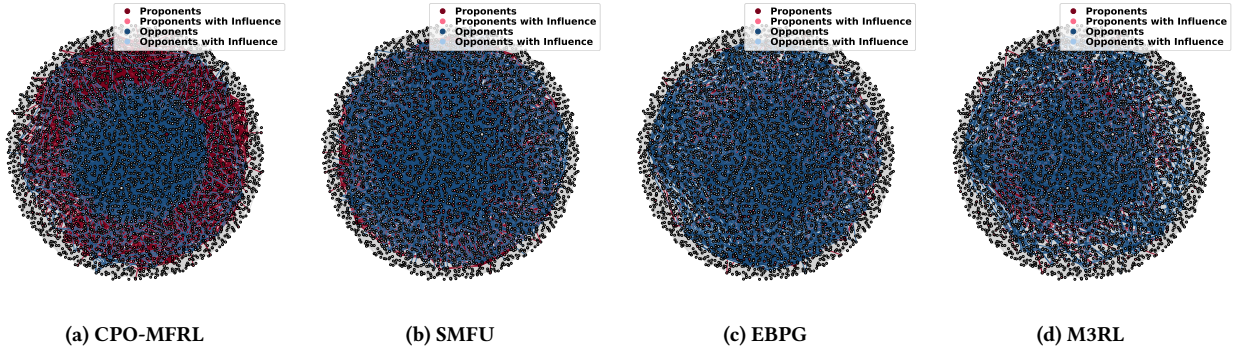


Figure 4: Visualization of information propagation achieved by CPO-MFRL and baselines on the Facebook dataset.

intricate structural patterns of the network data and learn superior, population-aware equilibrium strategies for influence propagation.

Comparison of Exploitability: Figure 3 compares the exploitability achieved by CPO-MFRL and baselines across 4 datasets. The results demonstrate that the proposed CPO-MFRL algorithm consistently achieves lower asymptotic exploitability compared to baselines, with reductions of approximately 51.58%, 49.84%, 52.78%, and 47.37% for 4 datasets respectively. The consistent decrease in exploitability demonstrates CPO-MFRL’s effective convergence toward Nash Equilibrium, confirming its robustness across diverse network environments.

Visualization of Seeds and Propagation: Figure 4 illustrates seed selection and activation patterns on the Facebook dataset, showing CPO-MFRL achieves broader influence coverage in DPPO-MFG than baselines. Notably, under extensive opponent propagation (light blue), the proponent population (light red) maintains wider influence distribution, demonstrating CPO-MFRL’s strategic prioritization of high-impact nodes under competitive pressure.

Ablation Experiments: The result of ablation study is detailed in Appendix E. Our method consistently outperforms CPO-MFRL-DDPG and CPO-MFRL without Auto-Encoder in seed coverage rate across budget levels. The enhanced stability further validates the contribution of each algorithmic module.

6 Conclusions

This paper proposes DPPO-MFG, a novel framework for modeling competitive information propagation in social networks. It captures key real-world dynamics, including user heterogeneity and adversarial interactions between two populations. Theoretically, we prove the existence of a unique Nash equilibrium under bounded rationality. Computationally, we develop CPO-MFRL, a scalable deep reinforcement learning algorithm employing graph neural networks for end-to-end equilibrium learning. Experiments across multiple datasets demonstrate the framework’s robustness and superior performance in stability and influence maximization over existing baselines.

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References

- [1] Yves Achdou, Guy Barles, Hitoshi Ishii, Grigory L Litvinov, and Yves Achdou. 2013. Finite difference methods for mean field games. *Hamilton-Jacobi Equations: Approximations, Numerical Analysis and Applications: Cetraro, Italy 2011, Editors: Paola Loreti, Nicoletta Anna Tchou* (2013), 1–47.
- [2] Yves Achdou and Italo Capuzzo-Dolcetta. 2010. Mean field games: numerical methods. *SIAM J. Numer. Anal.* 48, 3 (2010), 1136–1162.
- [3] Khurshed Ali, Chih-Yu Wang, and Yi-Shin Chen. 2018. Boosting reinforcement learning in competitive influence maximization with transfer learning. In *2018 IEEE/WIC/ACM International Conference on Web Intelligence (WI)*. IEEE, 395–400.
- [4] Khurshed Ali, Chih-Yu Wang, and Yi-Shin Chen. 2022. Leveraging transfer learning in reinforcement learning to tackle competitive influence maximization. *Knowledge and Information Systems* 64, 8 (2022), 2059–2090.
- [5] Khurshed Ali, Chih-Yu Wang, Mi-Yen Yeh, and Yi-Shin Chen. 2020. Addressing competitive influence maximization on unknown social network with deep reinforcement learning. In *2020 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*. IEEE, 196–203.
- [6] Khurshed Ali, Chih-Yu Wang, Mi-Yen Yeh, Cheng-Te Li, and Yi-Shin Chen. 2021. NEDRL-CIM: Network embedding meets deep reinforcement learning to tackle competitive influence maximization on evolving social networks. In *2021 IEEE 8th International Conference on Data Science and Advanced Analytics (DSAA)*. IEEE, 1–9.
- [7] D. Bauso, H. Tembine, and T. Başar. 2016. Opinion Dynamics in Social Networks through Mean-Field Games. *SIAM J. Control Optim.* 54, 6 (2016), 3225–3257.
- [8] Shishir Bharathi, David Kempe, and Mahyar Salek. 2007. Competitive influence maximization in social networks. In *Internet and Network Economics: Third International Workshop, WINE 2007, San Diego, CA, USA, December 12–14, 2007. Proceedings* 3. Springer, 306–311.
- [9] Gokce Dayanikli and Mathieu Lauriere. 2023. Multi-population Mean Field Games with Multiple Major Players: Application to Carbon Emission Regulations. arXiv:2309.16477 [math.OC] <https://arxiv.org/abs/2309.16477>
- [10] Christian G Fink, Kelly Fullin, Guillermo Gutierrez, Nathan Omodt, Sydney Zinnecker, Gina Sprint, and Sean McCulloch. 2023. A centrality measure for quantifying spread on weighted, directed networks. *Physica A* (2023).
- [11] Xin Guo, Anran Hu, Renyuan Xu, and Junzi Zhang. 2019. Learning mean-field games. *Advances in Neural Information Processing Systems* 32 (2019).
- [12] Qiang He, Xingwei Wang, Yong Zhao, Bo Yi, Xijia Lu, Mingzhou Yang, and Min Huang. 2021. Reinforcement-learning-based competitive opinion maximization approach in signed social networks. *IEEE Transactions on Computational Social Systems* 9, 5 (2021), 1505–1514.
- [13] Srijan Kumar, Bryan Hooi, Disha Makhija, Mohit Kumar, Christos Faloutsos, and VS Subrahmanian. 2018. Rev2: Fraudulent user prediction in rating platforms. In *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining*. ACM, 333–341.
- [14] Srijan Kumar, Francesca Spezzano, VS Subrahmanian, and Christos Faloutsos. 2016. Edge weight prediction in weighted signed networks. In *Data Mining (ICDM), 2016 IEEE 16th International Conference on*. IEEE, 221–230.
- [15] Mathieu Laurière, Sarah Perrin, Sertan Girgin, Paul Muller, Ayush Jain, Theophile Cabannes, Georgios Piliouras, Julien Pérolat, Romuald Élie, Olivier Pietquin, and Matthieu Geist. 2022. Scalable Deep Reinforcement Learning Algorithms for Mean Field Games. arXiv:2203.11973 [cs.LG] <https://arxiv.org/abs/2203.11973>
- [16] Pengdeng Li, Runsheng Yu, Xinrun Wang, and Bo An. 2024. Transition-Informed Reinforcement Learning for Large-Scale Stackelberg Mean-Field Games. In *Thirty-Eighth AAAI Conference on Artificial Intelligence*, Michael J. Wooldridge, Jennifer G. Dy, and Sriraam Natarajan (Eds.). AAAI Press, 17469–17476.
- [17] Su-Chen Lin, Shou-De Lin, and Ming-Syan Chen. 2015. A learning-based framework to handle multi-round multi-party influence maximization on social networks. In *Proceedings of the 21th ACM SIGKDD international conference on knowledge discovery and data mining*. 695–704.
- [18] Julian McAuley and Jure Leskovec. 2012. Learning to discover social circles in ego networks. In *Proceedings of the 26th International Conference on Neural Information Processing Systems - Volume 1 (Lake Tahoe, Nevada) (NIPS'12)*. Curran Associates Inc., Red Hook, NY, USA, 539–547.
- [19] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Alex Graves, Ioannis Antonoglou, Daan Wierstra, and Martin Riedmiller. 2013. Playing atari with deep reinforcement learning. arXiv preprint arXiv:1312.5602 (2013).
- [20] Alessandro Nardio, Alberto Tarable, Carla-Fabiana Chiasserini, and Emilio Leonardi. 2018. Belief Dynamics in Social Networks: A Fluid-Based Analysis. *IEEE Transactions on Network Science and Engineering* 5, 4 (2018), 276–287.
- [21] Georgios Papoudakis, Filippos Christianos, and Stefano V. Albrecht. 2020. Opponent Modelling with Local Information Variational Autoencoders. CoRR abs/2006.09447 (2020). arXiv:2006.09447 <https://arxiv.org/abs/2006.09447>
- [22] Luigi De Pascale and Luca Nenna. 2024. A variational formulation of a Multi-Population Mean Field Games with non-local interactions. arXiv:2408.03118 [math.AP] <https://arxiv.org/abs/2408.03118>
- [23] Sarah Perrin, Mathieu Laurière, Julien Pérolat, Matthieu Geist, Romuald Élie, and Olivier Pietquin. 2021. Mean Field Games Flock! The Reinforcement Learning Way. arXiv:2105.07933 [cs.MA] <https://arxiv.org/abs/2105.07933>
- [24] Zhenyu Shi*, Runsheng Yu*, Xinrun Wang*, Rundong Wang, Youzhi Zhang, Hanjiang Lai, and Bo An. 2020. Learning Expensive Coordination: An Event-Based Deep RL Approach. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=rYeG924twB>
- [25] Tianmin Shu and Yuandong Tian. 2019. M³RL: Mind-aware Multi-agent Management Reinforcement Learning. arXiv:1810.00147 [cs.AI] <https://arxiv.org/abs/1810.00147>
- [26] Jayakumar Subramanian and Aditya Mahajan. 2019. Reinforcement learning in stationary mean-field games. In *Proceedings of the 18th International Conference on Autonomous Agents and Multiagent Systems*. 251–259.
- [27] Sriram Ganapathi Subramanian, Matthew E. Taylor, Mark Crowley, and Pascal Poupart. 2021. Partially Observable Mean Field Reinforcement Learning. arXiv:2012.15791 [cs.MA] <https://arxiv.org/abs/2012.15791>
- [28] Hamidou Tembine, Dario Bauso, and Tamer Başar. 2013. Robust linear quadratic mean-field games in crowd-seeking social networks. In *52nd IEEE Conference on Decision and Control*. 3134–3139.
- [29] Andrew Wang, Andrew C. Li, Torny Q. Klassen, Rodrigo Toro Icarte, and Sheila A. McIlraith. 2023. Learning Belief Representations for Partially Observable Deep RL. In *International Conference on Machine Learning, ICML 2023, 23–29 July 2023, Honolulu, Hawaii, USA (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, 35970–35988. <https://proceedings.mlr.press/v202/wang23p.html>
- [30] Guofang Wang, Ziming Li, Wang Yao, and Sikai Xia. 2022. A Multi-Population Mean-Field Game Approach for Large-Scale Agents Cooperative Attack-Defense Evolution in High-Dimensional Environments. *Mathematics* 10, 21 (2022). doi:10.3390/math10214075
- [31] Tzu-Hsin Yang, Hao-Shang Ma, and Jen-Wei Huang. 2018. DNA: General deterministic network adaptive framework for multi-round multi-party influence maximization. In *2018 IEEE 5th International Conference on Data Science and Advanced Analytics (DSAA)*. IEEE, 273–282.
- [32] Yaodong Yang, Rui Luo, Minne Li, Ming Zhou, Weinan Zhang, and Jun Wang. 2018. Mean field multi-agent reinforcement learning. In *International Conference on Machine Learning*. PMLR, 5571–5580.
- [33] Chao Yu, Xinyi Yang, Jiaxuan Gao, Jiayu Chen, Yunfei Li, Jijia Liu, Yunfei Xiang, Ruixin Huang, Huazhong Yang, Yi Wu, and Yu Wang. 2023. Asynchronous Multi-Agent Reinforcement Learning for Efficient Real-Time Multi-Robot Cooperative Exploration. arXiv:2301.03398 [cs.RO] <https://arxiv.org/abs/2301.03398>
- [34] Zejian Zhou, Yuzhu Zhang, and Hao Xu. 2021. Reinforcement learning-based decentralized optimal control for large-scale multi-agent system by using neural networks and discrete-time mean field games. In *2021 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 1–6.

8 Appendix

A Dual-Population Policy-Aware Model

In this section, We extend PAM to the dual-population setting and denote the model by $\tilde{M} := \{\tilde{m}^w, S^w, A^w, T, \tilde{\mathcal{P}}^w, \tilde{r}^w\}_{w \in [W]}$, where S^w, A^w, T, m^w follow the dual-group MF-MDP setting. The transition $\tilde{\mathcal{P}}_t^w : S_t^w \times A_t^w \times \Pi \rightarrow \Delta(S_{t+1})$ and reward $\tilde{r}_t^w : S_t^w \times A_t^w \times \Pi \rightarrow [0, \frac{1}{T}]$ depend on the joint Markov policy $\pi := \{\pi^w\}_{w \in [W]}$.

Given a reference policy $\pi \in \Pi$, for any $\tilde{\pi} := \{\tilde{\pi}^w\}_{w \in [W]} \in \Pi$, we define the value function for population w , $\tilde{Q}_t^{w, \tilde{\pi}} : S_t^w \times A_t^w \times \Pi \rightarrow \mathbb{R}$ and $\tilde{V}_t^{w, \tilde{\pi}} : S_t^w \times \Pi \rightarrow \mathbb{R}$ in the following way:

$$\begin{aligned} \tilde{Q}_t^{w, \tilde{\pi}}(s_t^w, a_t^w, \tilde{a}_t^w; \pi) := & (1 - \alpha) \tilde{Q}_t^{w, \tilde{\pi}}(s_t^w, a_t^w, \tilde{a}_t^w; \pi) + \\ & \alpha [\tilde{r}_t^w(s_t^w, a_t^w, \pi) + \gamma v^w(s_{t+1}^w)] \end{aligned}$$

where $v^w(s_{t+1}^w) = \sum_{a_{t+1}^w} \pi^w \cdot \tilde{Q}_t^{w, \tilde{\pi}}(s_t^w, a_{t+1}^w, \tilde{a}_t^w; \pi)$. For partial observation, we estimate the mean action by Dirichlet sampling:

$$\tilde{a}_{i,t}^w \sim D^w(\theta; \eta + c); \tilde{a}_t^w = \frac{1}{U_{PO}} \sum_{i=1}^{U_{PO}} \tilde{a}_{i,t}^w \quad (4)$$

While in the absence of partially-observable, we define:

$$\begin{aligned} \tilde{Q}_t^{w, \tilde{\pi}}(s_t^w, a_t^w, \tilde{a}_t^w; \pi) := & (1 - \alpha) \tilde{Q}_t^{w, \tilde{\pi}}(s_t^w, a_t^w, \tilde{a}_t^w; \pi) + \\ & \alpha [\tilde{r}_t^w(s_t^w, a_t^w, \pi) + \gamma v^w(s_{t+1}^w)] \end{aligned}$$

where $v^w(s_{t+1}^w) = \sum_{a_{t+1}^w} \pi^w \cdot \tilde{Q}_t^{w, \tilde{\pi}}(s_t^w, a_{t+1}^w, \tilde{a}_t^w; \pi)$. Similarly, we take U samples from the Dirichlet distribution to estimate the mean action $\tilde{a}$. The formula is as follows:

$$\tilde{a}_{i,t}^w \sim D^w(\theta; \eta + c); \tilde{a}_t^w = \frac{1}{U} \sum_{i=1}^U \tilde{a}_{i,t}^w \quad (5)$$

Define the return functional:

$$\tilde{J}^w(\tilde{\pi}, \tilde{m}^w, \pi) = \mathbb{E}_{s_1^w \sim \mu_1^w, a_1^w \sim \tilde{\pi}_1^w(\cdot | s_1^w)} [\tilde{Q}_1^{w, \tilde{\pi}}(s_1^w, a_1^w, \tilde{a}_1^w; \pi)] \quad (6)$$

to be the expected return of agents in type w in model $\tilde{M}$ by executing $\tilde{\pi}$ given π as the reference policy.

Definition 4 (Nash Equilibrium in Dual-Population PAM). A policy π^{NE} is a Nash equilibrium if for all w and all $\tilde{\pi} \in \Pi$,

$$\tilde{J}^w(\tilde{\pi}, \tilde{m}^w, \pi^{NE}) \leq \tilde{J}^w(\pi^{NE}, \tilde{m}^w, \pi^{NE}). \quad (7)$$

A.1 Existence of Nash Equilibrium in DP-PAM

We consider the strict NE notion.

Definition 5 (Strict NE). Given a DP-PAM $\tilde{M}$ with transitions and rewards $\{\tilde{\mathcal{P}}^w, \tilde{r}^w\}_{w \in [W]}$, the policy π^{NE} is a strict NE of $\tilde{M}$ if and only if the following holds:

$$\pi_t^w(\cdot | s_t^w) \in \arg \max_{u \in \Delta(A^w)} \tilde{Q}_t^{w, \pi}(s_t^w, \cdot; \pi)^T u, \forall w \in [W]. \quad (8)$$

Note that this is a stronger notion than the NE defined in Definition 4, i.e. a strict NE is always a NE.

Lemma 1 (Strict NE as Fixed Point). Given a DP-PAM $\tilde{M}$ with transitions and rewards $\{\tilde{\mathcal{P}}^w, \tilde{r}^w\}_{w \in [W]}$, the policy π^{SNE} is a strict NE of $\tilde{M}$ if and only if the following holds:

$$\Gamma_{\tilde{M}}^{SNE}(\pi^{SNE}) = \pi^{SNE},$$

where $\Gamma_{\tilde{M}}^{SNE}(\pi) := \{\tilde{\pi} := \{\tilde{\pi}^w\}_{w \in [W], t \in [T]} | \forall w, s_t^w, \tilde{\pi}_t^w(\cdot | s_t^w) := \arg \max_{u \in \Delta(A^w)} \tilde{Q}_t^{w, \pi}(s_t^w, \cdot; \pi)^T u - \|\pi_t^w(\cdot | s_t^w) - u\|_2^2\}$

Proof Sketch: Strict NE $\Rightarrow$ Fixed Point: By definition, strict NE implies each π^w is unique best response. The quadratic penalty $-\|\pi_t^w - u\|_2^2$ is maximized at $u = \pi_t^w$, so $\Gamma_{\tilde{M}}^{SNE}(\pi) = \pi$. **Fixed Point $\Rightarrow$ Strict NE:** Fixed point condition implies first-order optimality. The penalty's gradient vanishes, reducing to the best-response condition. Thus π is a strict NE.

Definition 6 (Policy Distance). Given two policies $\pi := \{\pi_t^w\}_{t \in [T], w \in [W]}$ and $\tilde{\pi} := \{\tilde{\pi}_t^w\}_{t \in [T], w \in [W]}$, we define:

$$d_{\infty,1}(\pi, \tilde{\pi}) := \max_{w \in [W], t \in [T], s_t^w \in S_t^w} \|\pi_t^w(\cdot | s_t^w) - \tilde{\pi}_t^w(\cdot | s_t^w)\|_1.$$

Condition 1. Given a model function $\tilde{M} := \{(\tilde{\mathcal{P}}_t^w, \tilde{r}_t^w)\}_{t \in [T]}$, for any fixed $w \in [W]$, $t \in [T]$, s_{t+1}^w, s_t^w, a_t^w , and any $\pi, \tilde{\pi}$, $\tilde{\mathcal{P}}_t^w(s_{t+1}^w | s_t^w, a_t^w, \pi)$ and $\tilde{r}_t^w(s_t^w, a_t^w, \pi)$ are continuous at π w.r.t. distance $d_{\infty,1}$.

Lemma 2 (Continuity of $\tilde{Q}$). Under Condition 1, for any $w \in [W]$, any s_t^w, a_t^w , and π , $\tilde{Q}_t^{w, \pi}(s_t^w, a_t^w, \pi)$ is continuous at π under $d_{\infty,1}$.

PROOF. The proof is obvious by noting that $\tilde{Q}_t^{w, \pi}(s_t^w, a_t^w, \pi)$ is a function resulting from finite multiplication and addition among $\tilde{\mathcal{P}}_t^w(\cdot | \cdot, \cdot, \pi)$ and $\tilde{r}_t^w(\cdot, \cdot, \pi)$ and π . $\square$

Lemma 3. Under Condition 1, the DP-PAM has at least one NE policy π^{NE} satisfying Definition 4

Proof Sketch: Continuity: $\Gamma_{\tilde{M}}^{SNE}$ is continuous (by Condition 1, Lemma 2, and continuity of inner argmax maps). **Fixed Point:** Policy space Π is closed; by Brouwer's theorem, $\Gamma_{\tilde{M}}^{SNE}$ has a fixed point. **Implication:** Fixed point is strict NE (Lemma 1), and strict NE $\subseteq$ NE, proving existence.

Comparing with Definition 4 and Definition 5, we know NE is a super-set of strict NE, which implies the existence of NE in the DP-PAM.

A.2 Existence of Nash Equilibrium in DPPO-MFG as Corollary

Conversion from Dual-Population MF-MDP to DP-PAM. Given a Dual-Population MF-MDP $\tilde{M}$, we can convert it to a DP-PAM sharing the same $\{m_0^w, S^w, A^w, T\}_{w \in [W]}$ with $\tilde{M}$, while the transition and reward functions of $\tilde{M}$ are defined by:

$$\begin{aligned} \forall w \in [W], \forall t \in [T], \tilde{\mathcal{P}}_t^w(\cdot | \cdot, \cdot, \pi) &:= \mathcal{P}_t^w(\cdot | \cdot, \cdot, m_t^\pi), \\ \tilde{r}_t^w(\cdot, \cdot, \pi) &:= r_t(\cdot, \cdot, m_t^\pi), \end{aligned} \quad (9)$$

where m_t^π is the density of agents in all types induced by policy π in model $\tilde{M}$ starting from $m_0^\pi = \mu_1$.

By Lemma 4, we know the DP-PAM converted from such dual-population MF-MDP satisfying Condition 1, and by Proposition 3, the DP-PAM has at least one NE.

A.3 Converting DP-PAM from Dual-Population MF-MDP

Lemma 4 (Lipschitz Continuity of DP-PAM). Given a dual-population MF-MDP $\tilde{M}$ satisfying the Lipschitz continuity condition in Assumption B, consider the DP-PAM $\tilde{M}$ converted from $\tilde{M}$ according to Equation (9), $\tilde{M}$ is also Lipschitz continuous that, $\forall w \in [W], t \in [T]$ and any $s_t^w \in S^w, a_t^w \in A^w$, let $d_{\infty,1}(\pi, \pi') = \sum_{t'=0}^t (1 + L_T)^{t-t'} = K$: $\|\tilde{\mathcal{P}}_t^w(\cdot | s_t^w, a_t^w, \pi) - \tilde{\mathcal{P}}_t^w(\cdot | s_t^w, a_t^w, \pi')\|_1 \leq L_T K$, $\|\tilde{r}_t^w(s_t^w, a_t^w, \pi) - \tilde{r}_t^w(s_t^w, a_t^w, \pi')\| \leq L_r K$.

Proof Sketch: Using Lemma 6, the difference in mean-field terms $\|m_t^\pi - m_t^{\pi'}\|_1$ and $\|\mu_{M,t}^\pi - \mu_{M,t}^{\pi'}\|_1$ is bounded by $d_{\infty,1}(\pi, \pi')W \sum_{t'=0}^{t-1} (1+L_T)^{t-t'}$. By the Lipschitz conditions on $\mathcal{P}$ and r , the differences in transitions and rewards are bounded by the above mean-field differences, hence by $d_{\infty,1}(\pi, \pi')$ times a constant depending on W, L_T, L_r and the time horizon.

A.4 Conversion from DP-MFG to MFG with Constrained Policy Space

Setting about DP-MFG and MF-MDP can be checked in Sec.4.2

A.4.1 Definition of transition/reward functions in the lifted MF-MDP. Given a mean-field $m_t := [\frac{m_t^P}{W}, \frac{m_t^O}{W}] \in \Delta(\bar{S})$ with $m_t^w \in \Delta(S^w)$, the converted MFG is defined (by definition of Π^Ψ , we consider the case when the state and action share the population index):

$$\begin{aligned} \bar{\mathcal{P}}_t(s_{t+1}^{\tilde{w}} \circ \tilde{w} | s_t^w \circ w, a_t^w \circ w, m_t) \\ = \begin{cases} \mathcal{P}_{M,t}^w(s_{t+1}^{\tilde{w}} | s_t^w, a_t^w, m_t^w), & \text{if } \tilde{w} = w \\ 0, & \text{otherwise.} \end{cases} \quad (10) \\ \bar{r}_t(s_t^{\tilde{w}} \circ w, a_t^w \circ w, m_t) = r_t^w(s_t^w, a_t^w, m_t^w). \end{aligned}$$

For the sake of rigor, we include the definition for the transition/reward functions on those $s^w \circ w$ and $a^{\tilde{w}} \circ \tilde{w}$ with $w \neq \tilde{w}$.

After specifying a policy $\pi \in \Pi^\Psi$, denote $\pi := \{\pi^w\}_{w \in [W]}$ to be the DP-MFG policy that π corresponds to, we can verify that the state density $m_t^\pi \in \Delta(\bar{S})$ evolves according to:

$$\forall t \in [T]: \quad \bar{m}_t^\pi \leftarrow \left[\frac{m_t^{P,\pi}}{2}, \frac{m_t^{O,\pi}}{2} \right]. \quad (11)$$

where recall $m_t^{w,\pi}$ denotes the density of type w induced by π in model M . To see this, by induction,

$$\begin{aligned} \bar{m}_t^\pi(s^w \circ w) \\ = \sum_{s_{t-1}^w \in S^w, a_{t-1}^w \in A^w} \bar{m}_{t-1}^{w,\pi}(s_{t-1}^w \circ w) \pi(a_{t-1}^w \circ w | s_{t-1}^w \circ w). \quad (12) \\ \mathcal{P}_t^w(s_t^w | s_{t-1}^w, a_{t-1}^w, \bar{m}_{t-1}^\pi) = \frac{m_t^{w,\pi}(s^w)}{W} \end{aligned}$$

Intuitively, in the converted MFG, following a policy $\pi \in \Pi^\Psi$, if an agent starts from the initial state with index w , it will follow a trajectory as if it is generated in the original DP-MFG.

Definition 7 (Trajectory Sampling Model in DP-MFG). The learner can query the sampling model with an arbitrary policy $\pi := \{\pi^P, \pi^O\}$, a population index w and another policy $\tilde{\pi} := \{\tilde{\pi}^P, \tilde{\pi}^O\}$, and receive a trajectory by executing $\tilde{\pi}^w$ while the transition and reward functions are fixed by π , i.e. $\mathcal{P}_{M^*,t}^w(\cdot | \cdot, \cdot, m_{M^*,t}^\pi)$ and $r_t^w(\cdot, \cdot, m_{M^*,t}^\pi)$.

Moreover, for learning in the lifted MFGs, note that a sampling model in $\bar{M}^*$ as described in Definition 2.1 (need to add in the main article) can be implemented by Definition 7. To see this, given two policies $\pi, \tilde{\pi} \in \Pi^\Psi$, which correspond to $\pi := \{\pi^P, \pi^O\}$ and $\tilde{\pi} := \{\tilde{\pi}^P, \tilde{\pi}^O\}$, respectively, the trajectory can be generated by first uniformly sampling $w \in [W]$, and then sampling a trajectory with Definition 7 with π converted from π , type w and policy $\tilde{\pi}^w$.

Lemma 5. Given a DP-MFG model class M satisfying Assumption B, consider its converted MF-MDP model class $\bar{M}$. For any $M \in \bar{M}_{\text{MFG}}$, and any $\pi, \tilde{\pi} \in \Pi^\Psi$, we have $\forall t \in [T]$ that: $\|\mathcal{P}_t(\cdot | s_t, a_t, m_t^\pi) -$

$$\mathcal{P}_t(\cdot | s_t, a_t, m_t^{\tilde{\pi}})\|_1 \leq L_T \|m_t^\pi - m_t^{\tilde{\pi}}\|_1, |r_t(s_t, a_t, m_t^\pi) - r_t(s_t, a_t, m_t^{\tilde{\pi}})| \leq L_T \|m_t^\pi - m_t^{\tilde{\pi}}\|_1.$$

Proof Sketch: According to the definition in Appendix A.4.1, for those s_t, a_t with different population indices, their transition or reward differences will be 0. Therefore, we only need to consider the case when s_t, a_t share the population index. As explained in Equation (11), given $\pi, \tilde{\pi} \in \Pi^\Psi$, which correspond to $\pi, \tilde{\pi} \in \Pi$, respectively, we have: $\|m_t^\pi - m_t^{\tilde{\pi}}\|_1 = \frac{1}{W} \sum_{w \in [W]} \|m_t^{w,\pi} - m_t^{w,\tilde{\pi}}\|_1$. Combining with Assumption B, we finish the proof.

Lemma 6 (Density Difference). Given arbitrary dual-population MF-MDPs M and M' , and two arbitrary policies π and π' , for any $t \in [T]$:

$$\begin{aligned} \|m_{M,t}^\pi - m_{M',t}^{\pi'}\|_1 &\leq \|m_{M,t-1}^\pi - m_{M',t-1}^{\pi'}\|_1 + W \cdot d_{\infty,1}(\pi, \pi') \\ &+ \sum_{w \in [W]} \mathbb{E}_{\pi, M(\pi)} [\|\mathcal{P}_{M,t}^w - \mathcal{P}_{M',t}^w\|_1]. \end{aligned}$$

Proof Sketch: Decomposition: For each population w , decompose $\|m_{M,t}^{w,\pi} - m_{M',t}^{w,\pi'}\|_1$ into: Previous mean-field difference. Policy difference term $\leq d_{\infty,1}(\pi, \pi')$. Expected transition difference. **Triangle Inequality:** Apply triangle inequality to separate these three components. **Aggregation:** Sum over all W populations to get the final bound.

A.5 Constrained Nash Equilibrium

Given any $\pi \in \Pi^\Psi$, we denote its corresponding policy in DP-MFG by $\pi := \{\pi^P, \pi^O\}$ with $\pi^w : S^w \rightarrow \Delta(A^w)$, and $\pi(a_t^w \circ w | s_t^w \circ w) = \pi^w(a_t^w | s_t^w)$. Conversely, given any $\pi := \{\pi^P, \pi^O\}$, we can convert it to a policy in Π^Ψ , which we denote by π . For $\hat{\pi}_{Cstr}^{NE}$, we denote its correspondence in DP-MFG by $\hat{\pi}^{NE} := \{\hat{\pi}^{NE,P}, \hat{\pi}^{NE,O}\}$.

Note that, given any $\pi, \tilde{\pi} \in \Pi^\Psi$ and their correspondence $\pi := \{\pi^P, \pi^O\}$ and $\tilde{\pi} := \{\tilde{\pi}^P, \tilde{\pi}^O\}$, we have:

$$\begin{aligned} \bar{J}(\tilde{\pi}; \pi) &= \mathbb{E}_{\tilde{\pi}; M(\pi)} \left[\sum_{t=0}^T r_{M,t}(s_t, a_t, \bar{m}_t^\pi) \right] \\ &= \frac{1}{W} \sum_{w \in [W]} \mathbb{E}_{\tilde{\pi}^w; M(\pi)} \left[\sum_{t=0}^T r_t^w(s_t^w, a_t^w, m_t^\pi) \right], \end{aligned}$$

Consider the case when $\pi = \hat{\pi}^{NE}$ and $\tilde{\pi}^{\tilde{w}} = \hat{\pi}^{NE, \tilde{w}}$ for all $\tilde{w}$ except $\tilde{w} = w$, we have: $\epsilon \geq \bar{J}(\tilde{\pi}, m_{Cstr}^{NE}) - \bar{J}(\hat{\pi}_{Cstr}^{NE}, m_{Cstr}^{NE}) = \frac{1}{W} (J^w(\tilde{\pi}, \hat{\pi}^{NE}) - J^w(\hat{\pi}^{NE}, \hat{\pi}^{NE}))$. On the other hand, given an ϵ -approximate NE $\hat{\pi}^{NE}$ in M and its corresponding $\hat{\pi}_{Cstr}^{NE}$ in $\bar{M}$, for any $\pi \in \Pi^\Psi$, we have:

$$\begin{aligned} \bar{J}(\pi, m_{Cstr}^{NE}) - \bar{J}(\hat{\pi}_{Cstr}^{NE}, m_{Cstr}^{NE}) \\ = \frac{1}{W} \sum_{w \in [W]} \mathbb{E}_{\pi^w; M(\hat{\pi}^{NE})} \left[\sum_{t=1}^T r_t^w \right] - \mathbb{E}_{\hat{\pi}^{NE, w}; M(\hat{\pi}^{NE})} \left[\sum_{t=1}^T r_t^w \right] \quad (13) \\ = J^w(\tilde{\pi}, m_{Cstr}^{NE}) - J^w(\hat{\pi}^{NE}, m_{Cstr}^{NE}) \leq \epsilon, \end{aligned}$$

To upper bound the RHS, for each $w \in [W]$, we consider an arbitrary policy $\tilde{\pi}$ with $\tilde{\pi}^{\tilde{w}} = \hat{\pi}^{NE, \tilde{w}}$ for all $\tilde{w}$ except $\tilde{\pi}^w = \pi^w$, we should have:

By repeating for all $w \in [W]$, we complete the proof.

B Proof on Nash Equilibrium under Partial Observation

B.1 Limited Action Distance

Proof Sketch: Hoeffding's inequality: Actions are bounded in $[0, 1]$, so empirical mean $\tilde{a}$ concentrates around true mean $\bar{a}$. **Large samples:** N agents provide sufficient samples for Dirichlet distribution to approximate true mean. **Error bound:** Solving $2 \exp(-2nu^2) = \delta$ yields $u = \sqrt{\frac{1}{2n} \log \frac{2}{\delta}}$.

B.2 Limited Q-function Distance

Proof Sketch: Lipschitz Q-function: $|\tilde{Q}(\tilde{a}) - \tilde{Q}(\bar{a})| \leq M|\tilde{a} - \bar{a}|$. **Action concentration:** From Theorem X, each action component concentrates with bound $\sqrt{\frac{1}{2n} \log \frac{2}{\delta}}$. **Probability adjustment:** Due to sum-to-1 constraint, joint probability $\geq \delta^{L-1}$ for L components. **Final bound:** Combine to get $ML(\sqrt{\frac{1}{2n} \log \frac{2}{\delta}})^2$ bound.

B.3 Existence of Constrained NE Policy under Partial Observation

We process the results obtained from Proposition 4:

$$\begin{aligned} & \mathbb{E}_{s_1^w \sim \mu_1^w, a_1^w \sim \tilde{\pi}_1^w(\cdot | s_1^w)} \\ & [|\tilde{Q}_t^{w, \tilde{\pi}}(s_t^w, a_t^w, \tilde{a}_t^w; \pi^{NE}) - \tilde{Q}_t^{w, \tilde{\pi}}(s_t^w, a_t^w, \tilde{a}_t^w; \pi^{NE})|] \\ & = |J_{PO}^w(\tilde{\pi}, m^{\tilde{\pi}^{NE}}) - J^w(\tilde{\pi}, m^{\tilde{\pi}^{NE}})| \leq ML \log \frac{2}{\delta} \frac{1}{2n} \end{aligned} \quad (14)$$

Then we combine Eq.14:

$$|J_{PO}^w(\tilde{\pi}, m^{\tilde{\pi}^{NE}}) - J^w(\tilde{\pi}^{NE}, m^{\tilde{\pi}^{NE}})| \leq ML \log \frac{2}{\delta} \frac{1}{2n} + \epsilon \quad (15)$$

As implied by Eq.15 when $ML \log \frac{2}{\delta} \frac{1}{2n} + \epsilon \rightarrow 0$, any MF-MDP $\bar{M}$ converted from an DP-MFG M with NE(s) should have at least one constrained NE. Therefore, under the setting of partial observation in dual population mean field game, we can guarantee any model in the converted function class $\bar{M}$ has at least one constrained NE.

C Details about Datasets

The detailed information about the datasets is listed in the Table ??.

The US Congress on Twitter [10]: This dataset represents the Twitter interaction network for the 117th United States Congress. This dataset comprises 475 nodes and 13,289 edges, with an average node degree of 21.52.

Bitcoin trust weighted signed network [14][13]: This is who-trusts-whom network of people who trade using Bitcoin on a platform called Bitcoin Alpha and Bitcoin OTC. Bitcoin Alpha consists of 3,783 nodes and 24,186 edges, with an average node degree of 3.73. Bitcoin OTC consists of 5,881 nodes and 35,592 edges, with an average node degree of 3.65.

Facebook [18]: This directed networks contains Facebook user-user friendships. Each node is associated with features that reflect user profiles, which have been anonymized to protect user privacy. This dataset comprises 4,039 nodes and 88,234 edges, with an average node degree of 21.85.

D Details about the Hyper-parameters

The reward weights for selecting seed nodes are set to $\alpha = 0.5$ and $\beta = 0.5$. In the context of information propagation, the maximum hop count for information dissemination is set to 2. In the CPO-MFRL agent configuration, the time discount factor for total reward is set to $\gamma = 0.99$, and the target network delay is set to 2. For the GAE model, the dimension of the embedding vector is set to 8 and both the encoder and decoder are implemented using two-layer GAT networks, with the weight λ for the $\mathcal{L}^T$ component of the total loss set to 0.1. The actor networks and critic networks are optimized using the Adam optimizer with a learning rate of 10^{-3} and 10^{-4} , respectively. We deploy the max capacity of the replay buffer to 10000 and select 128 samples as one batch for training. The experiments are conducted on an Ubuntu 20.04.3 LTS platform equipped with four NVIDIA GeForce RTX 4090 TI GPUs.

E Ablation Experiments

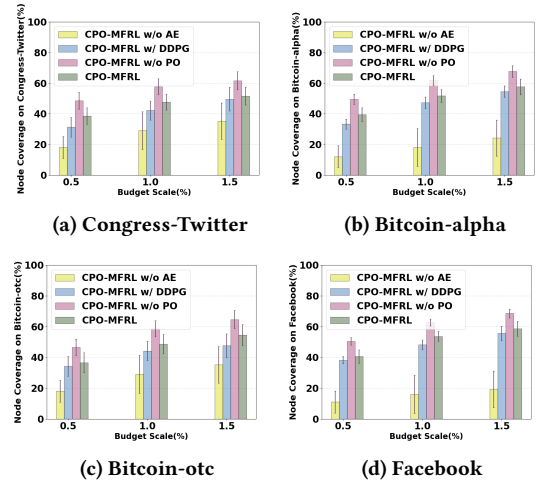


Figure 5: Comparison with ablation methods.

We further conduct ablation experiments to evaluate the efficacy of designing elements of CPO-MFRL on 4 different datasets. The experimental results are summarized in Figure 5. CPO-MFRL-DDPG replaces our DRL algorithm with DDPG. In the experimental results, CPO-MFRL-DDPG performs 4.17% lower than CPO-MFRL and exhibits 81.73% less stability on average. This can be attributed to the superior and stable outcomes resulting from the advantages of the dual critic networks and the network parameter update strategy. CPO-MFRL without Auto-Encoder performs 72.94% lower and is 252.2% less stable than CPO-MFRL on average, which indicates greater strategy robustness. The stability can be quantified by the standard deviation of the exploitability metric, which reflects the oscillation degree of strategies influencing node coverage in adversarial settings. This discrepancy is because embedding significantly enhances the understanding of the social network while reducing the randomness in seed node selection. CPO-MFRL without partial observability achieves 10.14% higher node coverage than the standard CPO-MFRL, validating the method's efficacy.